WTIM — Workflow Topology Integrity Module

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Canonical Language: English (UCL™)

Canonical Definition

Workflow Topology Integrity Module defines the governance conditions under which orchestrated workflows are formed, sequenced, connected, expanded, looped, or modified without losing structural validity, execution coherence, or governance control.

A system satisfies WTIM only if:

- workflow topology is explicitly defined or governably generated
- sequencing logic remains bounded and interpretable
- recursive, branching, or dynamic structures remain constrained
- execution paths do not expand beyond governed topology conditions
- workflow formation does not bypass authority, runtime, integrity, or ethical restrictions

A system that generates or changes workflow topology without governed integrity conditions does not satisfy WTIM.

Module Function

WTIM defines the structural integrity boundary of orchestration topology.

It ensures that workflow structure is not treated as a neutral technical arrangement, but as a governance object that determines how distributed execution becomes possible, limited, or invalid.

The module applies wherever autonomous systems build, modify, branch, recurse, reorder, or dynamically reconfigure execution pathways across agents, models, tools, or runtimes.

Minimum Implementation Framework (MIF)

Step 1 — Define Workflow Topology Conditions

The organization must define what types of workflow structures are valid.

Minimum requirement:

- allowed topology forms are explicit
- invalid or prohibited topology patterns are identifiable
- topology is not left structurally undefined

Step 2 — Define Sequencing Logic

The system must define how execution sequence is formed.

Minimum requirement:

- sequencing conditions are explicit
- dependencies between steps are visible
- hidden or arbitrary reordering is excluded from valid execution

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Step 3 — Define Branching and Recursion Boundaries

The system must define how branching, looping, recursion, or dynamic expansion is governed.

Minimum requirement:

- recursion boundaries are explicit
- branching conditions are identifiable
- self-expanding workflow behavior is not treated as valid by default

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Step 4 — Define Topology Modification Rules

The system must define when workflow structure may change during operation.

Minimum requirement:

- modification conditions are explicit
- dynamic topology change remains bounded
- silent structural mutation is excluded from valid orchestration

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Step 5 — Preserve Topology Traceability

The system must preserve the structure and path of workflow formation.

Minimum requirement:

- topology path is traceable
- later review can reconstruct how execution was structured
- branching, modification, and recursion remain visible

Step 6 — Restrict Invalid Workflow Formation

The system must not be treated as valid if workflow topology is used to bypass governance boundaries, conceal execution spread, or create structurally uncontrolled orchestration paths.

Minimum requirement:

- invalid topology conditions are identifiable
- uncontrolled recursion or hidden branching is blocked
- workflow convenience does not override governance validity

Use Case 1 — Multi-Agent Workflow Generation

Scenario

A coordinating agent dynamically builds a workflow by selecting specialized agents, branching subtasks, and reordering execution based on task complexity or intermediate outputs.

Application

WTIM is used to ensure:

- valid topology conditions are defined
- sequencing remains bounded and interpretable
- recursion and branching stay within governed limits
- workflow generation does not silently expand into uncontrolled orchestration

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Result

The organization gains a stronger basis for governing multi-agent workflow formation without allowing dynamic topology to become structurally opaque or unstable.

Use Case 2 — Adaptive Runtime Orchestration Environment

Scenario

A runtime system changes execution pathways during operation by re-routing steps, inserting new branches, or looping tasks across multiple coordinated agents.

Application

WTIM is used to ensure:

- topology modification rules are explicit
- branching and recursion remain constrained
- workflow path remains traceable
- execution does not continue under structurally ungoverned topology expansion

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Result

The organization gains a structurally governed workflow layer instead of relying on dynamic orchestration shape as if it were automatically valid.

Canonical Closing Statement

If workflow topology expands beyond governed structure, orchestration integrity has not been achieved.

Related Documents

→ Orchestration Governance Layer (OGL™)

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Canonical Source

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